NEAL HUANG

Applied AI & Backend Engineering
ckhuangdev@gmail.com | github.com/ackneal | linkedin.com/in/ackneal

SUMMARY

Applied AI and backend engineer with 10+ years of experience building distributed systems, search infrastructure, and production AI systems. Combines hands-on system architecture and implementation with engineering leadership, spanning conversational agents, content intelligence pipelines, and high-throughput transactional backends across multi-brand platforms.

EXPERIENCE

AI Team Lead | FunNow Jul 2025 – Present

- Lead an applied AI team building multi-brand conversational agents using Python, LangGraph, and MongoDB session persistence for intent-classified venue discovery and booking.
- Designed a model-routing architecture with prompt caching to direct lower-complexity tasks to lower-latency models while reserving frontier models for composite reasoning, reducing inference latency and cost.
- Established an LLM evaluation harness with Promptfoo to detect regressions across prompt and model changes before release.
- Built distributed content intelligence pipelines with Argo Workflows to transform venue metadata and public reviews into structured attributes for semantic search, automated tagging, and recommendations.
- Built an MCP server exposing venue discovery, booking, and order-management operations as typed tools for partner agents and customer-support integrations.

Backend Lead | FunNow Jan 2024 – Jun 2025

- Led a seven-person backend engineering team with ownership of core system architecture, cross-functional delivery, and multi-brand platform consolidation.
- Restructured core backend domains around explicit bounded contexts and service contracts using Domain-Driven Design and Clean Architecture, establishing a consistent architecture across backend repositories.
- Architected a parallel inventory and reservation engine supporting tiered-discount dining bookings without modifying legacy inventory state machines.
- Designed a zero-disruption dual-authentication migration from Firebase Auth to native platform identity with account deduplication and constant-time password verification.

Backend Engineer | FunNow Sep 2020 – Dec 2023

- Owned backend engineering for the Growth domain, spanning venue discovery and performance-critical search infrastructure.
- Built a write-preferring distributed read-write lock in Go with Redis hashes and Lua scripts, featuring ownership validation and context-bound lease renewal across payment and order workflows.
- Designed a two-layer geospatial discovery system using multi-precision GeoHash indexing in Elasticsearch and Redis GEO caching, with distributed single-flight cache population to protect the search cluster during concurrent cache misses.
- Executed a zero-downtime Elasticsearch 6-to-7 migration using asynchronous dual writes, shadow validation, and controlled query cutover, with automated index rotation and force-merge maintenance.

Advanced Backend Engineer | PIXNET Jul 2016 – Aug 2020

- Mentored three engineers on backend engineering practices while building backend services for social media, content discovery, and payments.
- Built a real-time member nickname autocomplete service using Elasticsearch Edge N-gram indexing for platform-wide tagging and mention discovery.
- Built a concurrent Open Graph crawler in Go and deployed it on GCP with horizontal autoscaling to power rich link previews.

Associate Project Manager | GigaFast Jul 2014 – Apr 2016

Managed project delivery, client communication, and cross-functional coordination before transitioning into backend engineering.

TECHNICAL SKILLS

Languages: Go, Python, PHP

AI & Agents: LangGraph, ReAct, MCP, LLM Evaluation (Promptfoo), Model Routing, Prompt Caching

Backend & Data: Distributed Systems, Elasticsearch, Redis, MySQL, MongoDB

Architecture & Platform: Domain-Driven Design, Clean Architecture, Argo Workflows, Docker, Kubernetes, AWS, GCP

EDUCATION

National Taipei University of Technology

B.S. in Industrial Engineering (Focus: Human Factors) | Taipei, Taiwan Sep 2009 – May 2013